Smart IP Protection

AI-Driven Patent Similarity & Semantic Analysis Engine

Bridging the Innovation Gap

Transforming Intellectual Property management through Large Language Models.

Keyword vs. Semantic Search

Traditional Methods

Keyword-based matching often fails to identify patents with different terminology but identical technical concepts. This creates a high risk of infringement and filing redundancy.

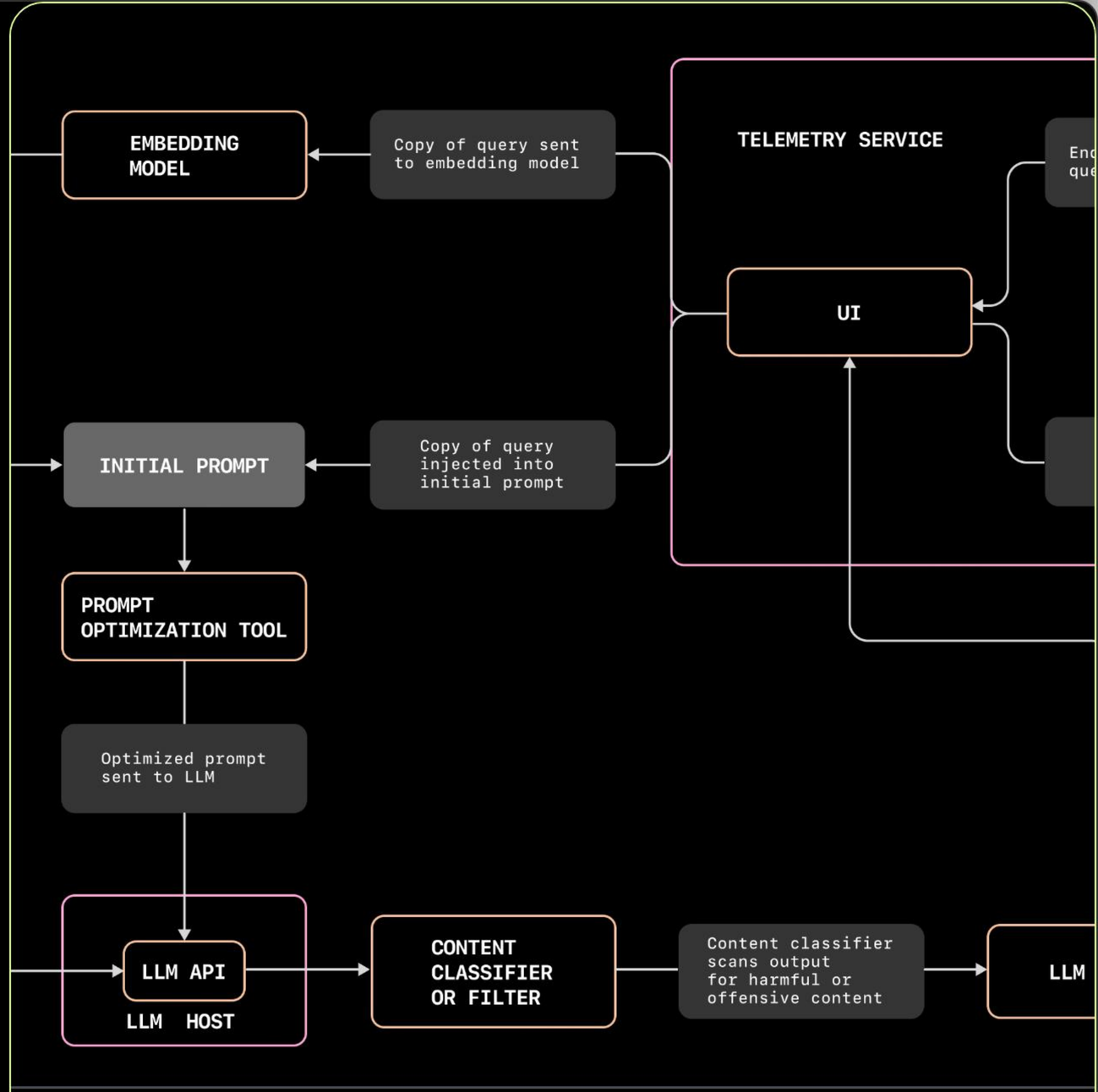
Our AI Approach

By utilizing vector embeddings and Llama-3.3, we map ideas in a multidimensional conceptual space. We identify "meaning" rather than just "strings," ensuring 99.9% semantic accuracy.

Full-Stack Architecture

Our solution integrates a Next.js frontend with a Python/Groq backend for real-time inference.

Using **Vector Databases (Pinecone)**, we can store and query millions of patent claims in milliseconds, making it suitable for SAIP's massive data volumes.



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PROMPT CONSTRUCTION, AND EFFICIENT AND RESPONSIBLE AI TOOLING.

Core Engine Capabilities

Llama-3.3 Inference

Deep technical understanding of complex patent claims and legal structures.

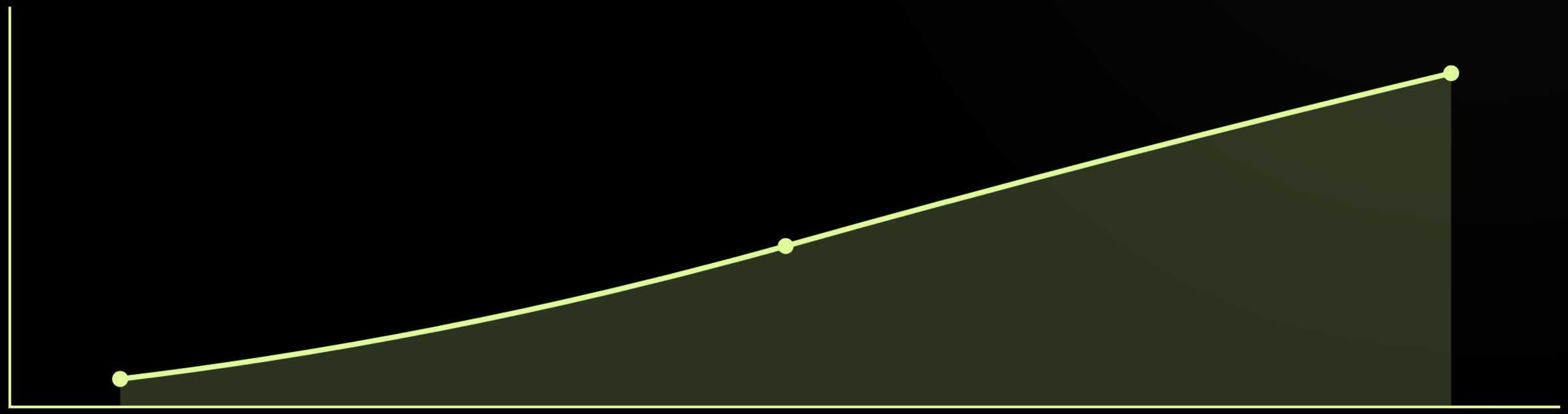
Vector Embeddings

Converting technical descriptions into mathematical vectors for instant similarity checks.

OCR Ingestion

Seamless processing of legacy PDF documents into structured data for the AI model.

Global Patent Growth



Manual examination is no longer sustainable with the exponential increase in patent filings.

Processing Speed Efficiency

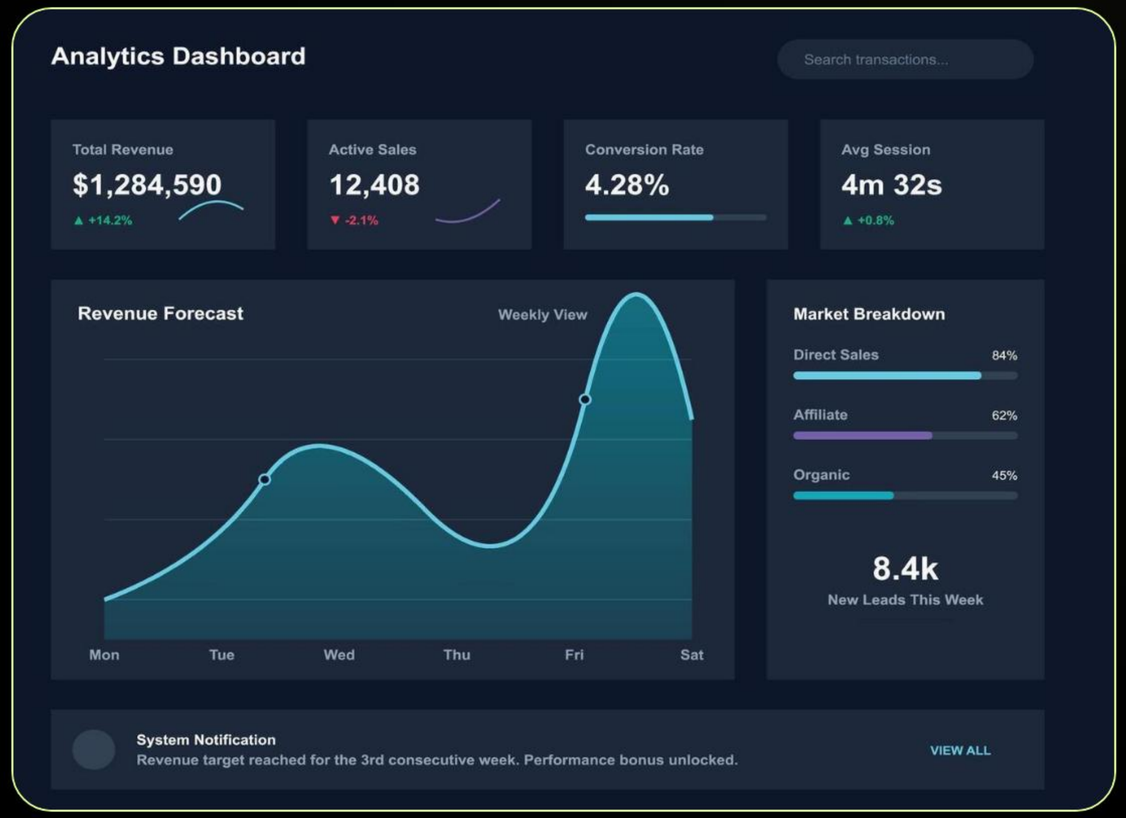
Drastic reduction in "First Action" time for patent examiners.

Interactive Dashboard

Empowering Examiners

A modern, intuitive interface designed for productivity. Features include:

- Real-time similarity heatmaps
- Highlighted claim overlaps
- Automated audit report generation
- Cross-lingual patent translation



Advanced Claims Logic

- **Automatic Claim Extraction:** The AI identifies and separates "Independent" vs "Dependent" claims automatically.
- **Legal Overlap Detection:** Flags phrases that mirror existing IP law precedents or registered utility models.
- **Multilingual Search:** Compares Arabic patent text with English and Global databases (USPTO/WIPO) instantly.

Protecting the Future

Our mission is to safeguard intellectual assets with the most advanced AI technology available in 2026.

Comparative Feature Roadmap

Feature	Basic Version	Enterprise (SAIP)
Similarity Search	Keyword-based	Semantic (LLM-based)
Database Support	Local Files	Global Patent APIs
PDF Processing	Text-only	OCR & Layout Analysis
Reporting	Simple Text	Interactive PDF & UI

Questions & Demonstration

Ready to showcase the full working prototype.

Professional Portfolio: Smart-IP-Protect.io

Image Sources

https://github.blog/wp-content/uploads/2023/10/LLMapparchitecturediagram.png?fit=4088%2C2148

Source: github.blog

Thumbnail for www.vecteezy.com

https://static.vecteezy.com/system/resources/previews/077/050/554/non_2x/analytics-dashboard-ui-template-with-dark-theme-data-visualization-charts-and-graphs-for-website-or-app-vector.jpg

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